Basic Linux Syntax

Question & Answer Cheat Sheet

1. Navigation & Files

Q1. How do you print the current working directory?

A:

```
pwd
```

Shows the full path of the directory you are in.

Q2. How do you list files in a directory?

A:

```
ls
```

Lists files and folders.

Q3. How do you list all files including hidden ones, with details?

A:

```
ls -la
```

-l gives long format, -a shows hidden files (names starting with .).

Q4. How do you change directory?

A:

```
cd /path/to/dir
```

Use cd .. to go up one level, cd ~ for home, cd - for previous directory.

Q5. How do you create a new directory?

A:

```
mkdir myfolder
```

Use mkdir -p a/b/c to create nested directories.

Q6. How do you create an empty file?

A:

```
touch file.txt
```

Also updates the timestamp if the file already exists.

Q7. How do you copy a file or directory?

A:

```
cp file.txt backup.txt
cp -r dir1 dir2
```

-r is needed to copy directories recursively.

Q8. How do you move or rename a file?

A:

```
mv old.txt new.txt
mv file.txt /tmp/
```

mv both moves and renames.

Q9. How do you delete a file or directory?

A:

```
rm file.txt
rm -r folder
```

Be careful: rm -rf deletes without confirmation and cannot be undone.

2. Viewing & Searching Text

Q10. How do you display a file's contents?

A:

```
cat file.txt
```

Use less file.txt for paging through large files.

Q11. How do you see the first or last lines of a file?

A:

```
head -n 10 file.txt
tail -n 10 file.txt
```

tail -f file.log follows a file live as it grows.

Q12. How do you search for text inside a file?

A:

```
grep "word" file.txt
```

-i ignores case, -r searches recursively, -n shows line numbers.

Q13. How do you find a file by name?

A:

```
find /home -name "file.txt"
```

find searches directories recursively.

Q14. How do you count lines, words and characters?

A:

```
wc file.txt
```

wc -l counts only lines.

Q15. How do you redirect output to a file?

A:

```
echo "hi" > file.txt
echo "more" >> file.txt
```

> overwrites the file, >> appends to it.

Q16. How do you pass one command's output to another?

A:

```
ls | grep ".txt"
```

The pipe | sends output of the left command as input to the right.

3. Permissions & Ownership

Q17. How do you view file permissions?

A:

```
ls -l
```

Output like -rwxr-xr-- shows owner, group, and others permissions.

Q18. How do you change permissions?

A:

```
chmod 755 script.sh
chmod +x script.sh
```

r=4, w=2, x=1. So 755 = rwx for owner, r-x for group and others.

Q19. How do you change a file's owner?

A:

```
sudo chown user:group file.txt
```

Use -R to apply recursively to a directory.

Q20. How do you run a command as administrator?

A:

```
sudo command
```

sudo runs a command with root privileges.

4. Processes & System

Q21. How do you list running processes?

A:

```
ps aux
```

top or htop shows live process usage.

Q22. How do you kill a process?

A:

```
kill 1234
kill -9 1234
```

Use the PID. -9 forces termination. killall name kills by name.

Q23. How do you run a command in the background?

A:

```
command &
```

Use jobs to list and fg to bring a job back to the foreground.

Q24. How do you check disk space?

A:

```
df -h
du -sh folder
```

df shows filesystem usage; du shows size of a folder. -h means human-readable.

Q25. How do you check memory usage?

A:

```
free -h
```

Shows total, used, and free RAM and swap.

Q26. How do you see who you are and system info?

A:

```
whoami
uname -a
```

whoami prints the current user; uname -a shows kernel and system info.

5. Packages, Networking & Misc

Q27. How do you install a package on Debian/Ubuntu?

A:

```
sudo apt update
sudo apt install package
```

On Fedora/RHEL use dnf; on Arch use pacman.

Q28. How do you check network connectivity?

A:

```
ping google.com
```

Press Ctrl+C to stop. Use ip a to see your IP addresses.

Q29. How do you download a file from the web?

A:

```
wget URL
curl -O URL
```

Both fetch files from a URL.

Q30. How do you read a command's manual?

A:

```
man ls
```

Press q to quit. Many commands also support --help.

Q31. How do you compress and extract files?

A:

```
tar -czvf a.tar.gz folder
tar -xzvf a.tar.gz
```

c=create, x=extract, z=gzip, v=verbose, f=file name.

Q32. How do you view command history?

A:

```
history
```

Use Ctrl+R to search previous commands, and !! to repeat the last one.

Q33. How do you clear the terminal?

A:

```
clear
```

Ctrl+L does the same.